Palestinian-origin Persons of Concern in Global Receiving Countries

Demography, distribution and host-country associations, 2020–2025

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1. Motivation and research questions

Palestinian displacement combines prolonged exile with renewed insecurity. Research on secondary displacement from Syria shows how repeated flight exposes gaps in international protection (Erakat, 2014). Gaza's humanitarian emergency makes accurate interpretation particularly important: OCHA reported approximately 1.9 million internally displaced people in December 2024 (United Nations Office for the Coordination of Humanitarian Affairs [OCHA], 2024). This crisis connects urgent needs in Gaza with longer-term protection concerns across receiving countries.

We examine global receiving countries during 2020–2025, asking how recorded persons of concern are distributed, how Palestinian-origin demographics and totals change, and whether host-country income and population correlate with their recorded numbers. The global overview provides context; the substantive models concern Palestinian origin only.

2. Dataset description

The supplied UNHCR demographic extract contains 35,480 observations and 20 variables (United Nations High Commissioner for Refugees [UNHCR], n.d.). Each row represents an origin–asylum–year combination. Year is integer; four country-name and ISO-code fields are character; fifteen population fields are integer counts, including age-by-sex categories, sex totals and Total. There are no missing cells or duplicate keys, but country-year coverage is uneven. Unreported country-years are treated as unavailable.

The extract reports persons of concern without separate population-type labels. Its accompanying Palestine footnotes specify UNHCR coverage. UNRWA operates separately in Jordan, Lebanon, Syria, Gaza and the West Bank (United Nations Relief and Works Agency for Palestine Refugees in the Near East [UNRWA], n.d.). Global country totals may also include domestic displacement.

The Palestinian subset contains 469 records across 93 receiving countries; 81 appear in 2025. World Bank GDP per capita in current US dollars and total population were retrieved for 2025 (World Bank, n.d.-a, n.d.-b). Each saved indicator table has 266 rows, including aggregates, and four variables: character ISO/name, integer year and numeric value. ISO matching retains 77 countries. Cuba, Lebanon, United Arab Emirates and Yemen lack GDP observations. Population is a numeric count; GDP is continuous. Matching selects individual countries.

```r
# Shared data for the four group contributions
library(dplyr)
library(tidyr)
library(ggplot2)
library(patchwork)
library(sf)

d <- read.csv(params$data_file, check.names = FALSE)
keys <- c("Year", "Country of Asylum ISO", "Country of Origin ISO")
stopifnot(nrow(d) == 35480L, ncol(d) == 20L,
          !anyNA(d), !anyDuplicated(d[keys]))
pal <- d %>% filter(`Country of Origin ISO` == "PSE")
pal25 <- pal %>%
  filter(Year == 2025) %>%
  transmute(country = `Country of Asylum`,
            iso3 = `Country of Asylum ISO`, total = Total)
country_year <- d %>%
  group_by(Year, `Country of Asylum ISO`) %>%
  summarise(Total = sum(Total), .groups = "drop")

# Part 4: use the saved World Bank 2025 indicator snapshots.
gdp <- read.csv(params$gdp_file)
pop <- read.csv(params$population_file)
stopifnot(!anyDuplicated(gdp$iso3[nzchar(gdp$iso3)]),
          !anyDuplicated(pop$iso3[nzchar(pop$iso3)]),
          all(gdp$year == 2025), all(pop$year == 2025))
merged <- pal25 %>%
  left_join(select(filter(gdp, nzchar(iso3)), iso3, gdp_pc),
            by = "iso3") %>%
  left_join(select(filter(pop, nzchar(iso3)),
                   iso3, host_population), by = "iso3")
excluded <- merged %>%
  filter(is.na(gdp_pc) | is.na(host_population))
analysis25 <- merged %>%
  filter(complete.cases(total, gdp_pc, host_population),
         gdp_pc > 0, host_population > 0)
stopifnot(nrow(pal) == 469L,
          n_distinct(pal$`Country of Asylum ISO`) == 93L,
          nrow(pal25) == 81L, nrow(analysis25) == 77L)
```

Table 1: Dataset dimensions

Dataset	Rows	Variables
UNHCR extract	35,480	20
Palestinian subset	469	20
World Bank GDP table	266	4
World Bank population table	266	4
Model 2 sample	77	5

Table 2: Countries excluded from Model 2 (NA means unavailable)

country	gdp_pc	host_population
Cuba	NA	10,937,203
Lebanon	NA	5,849,421
United Arab Emirates	NA	11,513,149
Yemen	NA	41,773,878

3. Quantitative summary

Across all original records, Total has mean 21,026.37, median 37, standard deviation 340,175.19, minimum 5 and maximum 23,880,052. Corresponding Palestinian statistics are 1,137.64, 46, 5,857.46, 5 and 70,022. Means greatly exceed medians, demonstrating strong right skew. Tables additionally summarise yearly subgroups, country-level totals and model covariates. Annual comparisons follow changing population stocks.

The Palestinian median increases from 32 in 2020 to 64 in 2025, while its mean decreases from 1,437.70 to 852.17. This divergence shows why a single average is insufficient: changes in the largest records can dominate the mean even when the typical reported country has a higher count.

```
summarise_counts <- function(x) {
  x %>% summarise(N=n(), Mean=mean(Total),
    Median=median(Total), SD=sd(Total),
    Minimum=min(Total), Maximum=max(Total), .groups="drop")
}
whole_summary <- bind_rows(
  summarise_counts(d) %>% mutate(Scope="All origins"),
  summarise_counts(pal) %>% mutate(Scope="Palestinian origin")
) %>% select(Scope,everything())
year_summary <- bind_rows(
  summarise_counts(group_by(d,Year)) %>% mutate(Scope="All origins"),
  summarise_counts(group_by(pal,Year)) %>% mutate(Scope="Palestinian")
) %>% mutate(Year=as.character(Year)) %>% select(Scope,Year,everything())
host_summary <- summarise_counts(group_by(country_year,Year)) %>%
  mutate(Year=as.character(Year))
covariate_summary <- analysis25 %>%
  select(total,gdp_pc,host_population) %>%
  pivot_longer(everything(),names_to="Variable",values_to="Total") %>%
  group_by(Variable) %>% summarise_counts() %>%
  mutate(Variable=recode(Variable,gdp_pc="GDP pc (US$)",
    host_population="Population",total="POC count"))
```

Table 3: Total across original records

Scope	N	Mean	Median	SD	Minimum	Maximum
All origins	35,480	21,026.37	37	340,175.19	5	23,880,052
Palestinian origin	469	1,137.64	46	5,857.46	5	70,022

Table 4: Total by origin scope and year

Scope	Year	N	Mean	Median	SD	Minimum	Maximum
All origins	2020	5,482	16,152.73	34.0	219,814.20	5	9,494,677
All origins	2021	5,571	17,176.93	36.0	221,463.94	5	8,007,891
All origins	2022	5,932	21,997.35	35.0	364,286.35	5	21,990,714
All origins	2023	6,036	23,620.64	36.0	391,729.71	5	23,289,649
All origins	2024	6,185	23,820.92	39.0	403,098.57	5	23,880,052
All origins	2025	6,274	22,534.08	39.0	367,948.10	5	22,047,185
Palestinian	2020	79	1,437.70	32.0	7,925.17	5	70,022
Palestinian	2021	78	1,465.83	41.0	7,979.31	5	70,021
Palestinian	2022	76	1,554.71	38.0	8,129.12	5	70,020
Palestinian	2023	78	722.18	39.5	2,247.66	5	15,265
Palestinian	2024	77	806.83	52.0	2,347.62	5	14,620
Palestinian	2025	81	852.17	64.0	2,329.53	5	15,142

Table 5: Global totals aggregated by receiving country and year

Year	N	Mean	Median	SD	Minimum	Maximum
2020	177	500,278.3	38,480	1,302,909	5	11,245,115
2021	179	534,595.8	36,070	1,316,181	5	9,880,716
2022	177	737,221.8	78,578	2,140,386	5	22,196,353
2023	177	805,503.8	88,714	2,327,365	5	23,466,521
2024	180	818,513.2	88,216	2,408,960	5	24,074,137
2025	177	798,750.4	93,032	2,249,469	5	22,223,539

Table 6: Model 2 complete-case sample (rounded to whole units)

Variable	N	Mean	Median	SD	Minimum	Maximum
GDP pc (US$)	77	26,300	13,862	30,917	627	147,252
Population	77	83,819,288	19,859,921	232,869,226	392,404	1,463,865,525
POC count	77	863	64	2,384	5	15,142

4. Visualisations

Figure 1 shows unequal global country totals: the median rises from 38,480 in 2020 to 93,032 in 2025. Figure 2 identifies a male-heavy recorded adult population. Other ages are displayed separately. Female and male totals sum to 60,724, leaving 8,302 of the overall 69,026 outside the supplied sex totals, indicating incomplete demographic coverage.

```r
# Figure 1: adapted from Part 2 (global distribution).
make_box <- function() {
  ggplot(country_year, aes(factor(Year), Total)) +
    geom_boxplot(fill = "#AAC7D2", colour = "#284B5B",
                 outlier.alpha = .3) +
    stat_summary(aes(group = 1), fun = median,
                 geom = "line", linetype = 2) +
    scale_y_log10(labels = scales::label_number(
      scale_cut = scales::cut_short_scale())) +
    labs(x = NULL, y = "Recorded persons") +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))
}
top3 <- country_year %>%
  group_by(Year) %>%
  slice_max(Total, n = 3, with_ties = FALSE) %>%
  arrange(Year, desc(Total)) %>%
  mutate(Rank = factor(row_number())) %>%
  ungroup()
p1 <- make_box() + labs(title = "A  Full distribution")
p2 <- make_box() + coord_cartesian(ylim = c(1000, 1000000)) +
  labs(title = "B  Central range", y = NULL)
p3 <- ggplot(top3, aes(Year, Total, colour = Rank, group = Rank)) +
  geom_line() + geom_point() +
  scale_x_continuous(breaks = 2020:2025) +
  scale_y_continuous(labels = scales::label_number(
    scale = 1e-6, suffix = "M")) +
  labs(title = "C  Three largest totals", x = NULL, y = NULL) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
p1 + p2 + p3
```

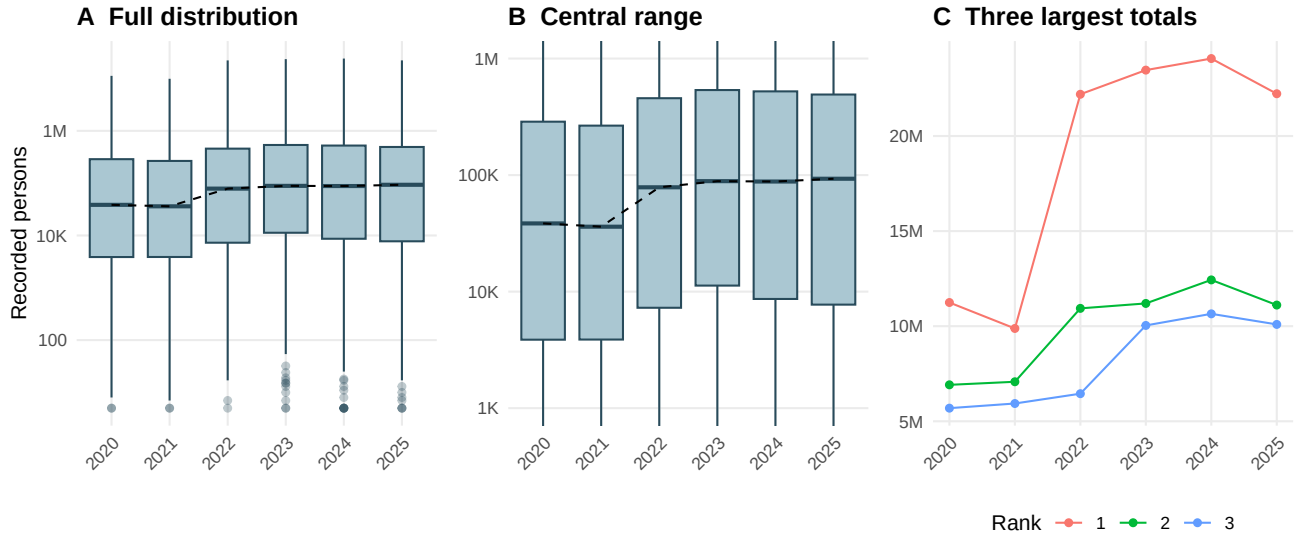


Figure 1: Global receiving-country distributions. Left and centre use log scales; right tracks each year's three largest totals.

```r
# Figure 2: adapted from Part 1 (demographic analysis).
# Keep the original Other category separate from 60+.
age <- pal %>%
  filter(Year == 2025) %>%
  summarise(across(
    matches("^(Female|Male) (0|5|12|18|60|Other)"), sum)) %>%
  pivot_longer(everything(), names_to = c("Sex", "Age"),
               names_pattern = "^(Female|Male) (.*)$",
               values_to = "Count") %>%
  mutate(
    Age = factor(Age, levels = c("0 - 4", "5 - 11", "12 - 17",
                                 "18 - 59", "60+", "Other")),
    Signed = if_else(Sex == "Female", -Count, Count)
  )
ggplot(age, aes(Signed, Age, fill = Sex)) + geom_col() +
  scale_x_continuous(labels = function(x) scales::comma(abs(x))) +
  scale_fill_manual(values = c(Female = "#B66942", Male = "#287F8A")) +
  labs(x = "Recorded persons", y = "Age category")
```

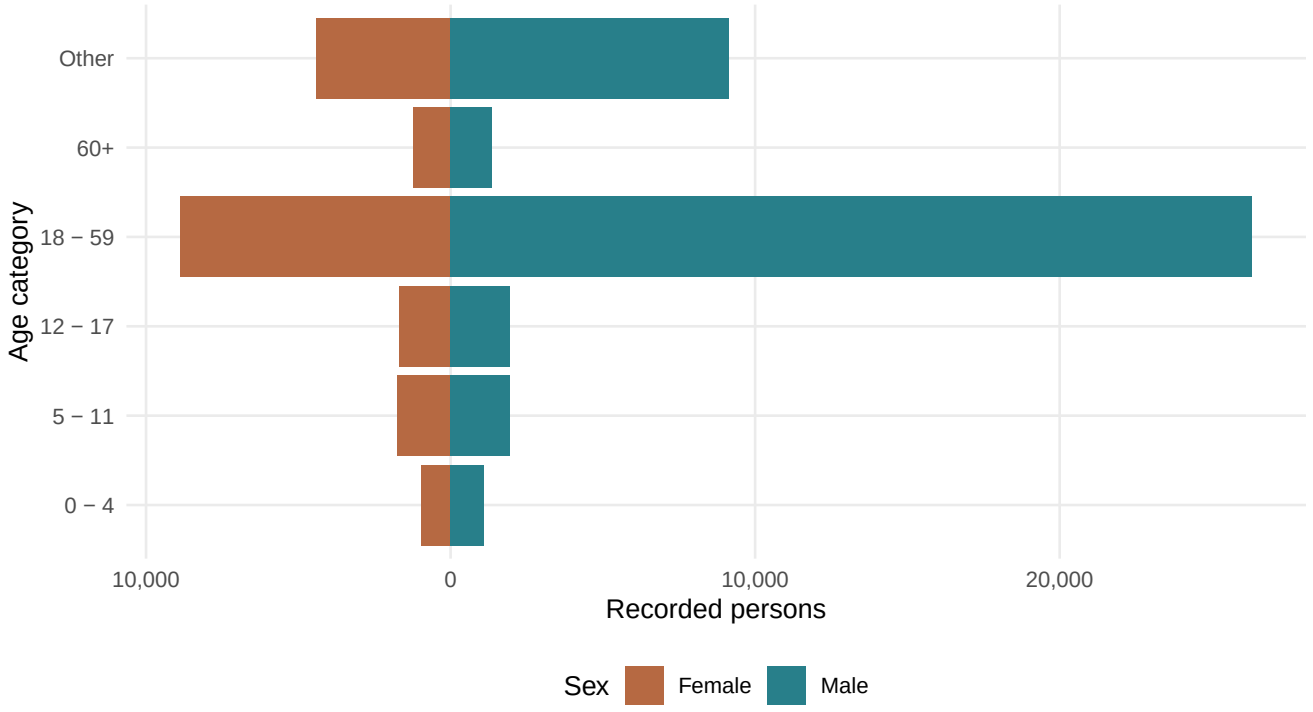


Figure 2: Palestinian-origin age and sex counts in 2025. Other is the source age category; 8,302 are outside the reported sex totals.

Figure 3 separates reporting discontinuity from apparent change. Recorded totals fall 52.3% between 2022 and 2023, whereas excluding Egypt from both years produces a 17.0% increase. Egypt's annual counts of approximately 70,000 disappear after 2022. The report documents a downward revision of its Palestinian estimate (UNHCR, 2024, p. 8). Excluding Egypt throughout yields 58.5% growth over 2020–2025, although other coverage changes remain possible.

```r
# Figure 3: adapted from Part 3 (population trends).
annual <- pal %>%
  group_by(Year) %>%
  summarise(
    All = sum(Total),
    `Excluding Egypt` = sum(Total[`Country of Asylum ISO` != "EGY"]),
    .groups = "drop"
  )
long_annual <- annual %>%
  pivot_longer(-Year, names_to = "Series", values_to = "Total")
ggplot(long_annual, aes(Year, Total, colour = Series,
                        linetype = Series)) +
  geom_line(linewidth = 1) + geom_point(size = 2) +
  scale_colour_manual(values = c("#287F8A", "#B66942")) +
  scale_x_continuous(breaks = 2020:2025) +
  scale_y_continuous(labels = scales::comma,
                     limits = c(0, 130000)) +
  labs(x = NULL, y = "Recorded persons",
       colour = NULL, linetype = NULL)
```

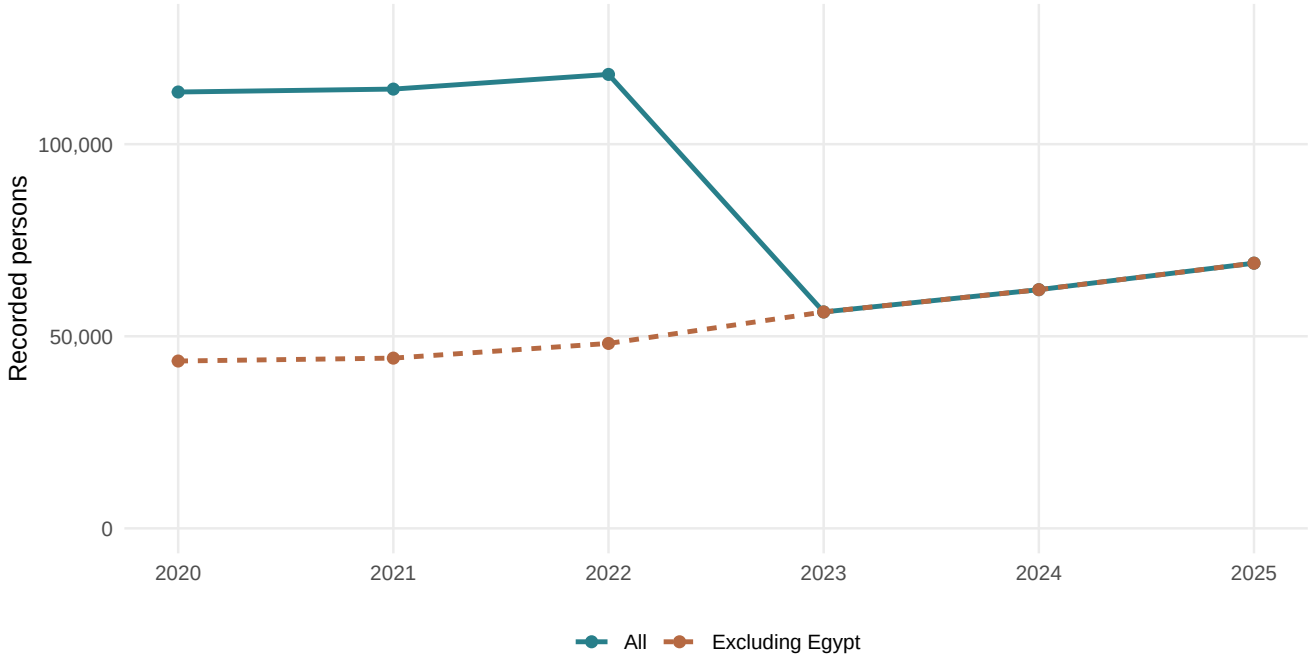


Figure 3: Palestinian-origin annual totals. Egypt is unreported from 2023 onwards, when the series overlap.

Table 7: Annual totals with and without Egypt

Year	All	Excluding Egypt
2020	113,578	43,556
2021	114,335	44,314
2022	118,158	48,138
2023	56,330	56,330
2024	62,126	62,126
2025	69,026	69,026

Figures 4–5 locate and rank the 2025 records. Belgium leads with 15,142, followed by Greece with 11,470 and Iraq with 7,477. Grey map areas indicate unmatched observations.

```
# Figure 4: adapted from the map in Part 1.
# Bundled geometry works offline; the fallback matches France/Norway.
world <- rnaturalearthdata::countries110
if (!inherits(world, "sf")) world <- sf::st_as_sf(world)
world$join_iso <- ifelse(world$iso_a3 == "-99",
                         world$adm0_a3, world$iso_a3)
map_data <- left_join(world, pal25, by = c("join_iso" = "iso3"))
ggplot(map_data) +
  geom_sf(aes(fill = total), colour = "white", linewidth = .12) +
  scale_fill_viridis_c(
    option = "magma", direction = -1, trans = "sqrt",
    limits = c(0, max(pal25$total)),
    breaks = c(0, 1000, 5000, 15000),
    na.value = "grey88", labels = scales::comma,
    guide = guide_colourbar(barwidth = grid::unit(9, "cm"),
                            barheight = grid::unit(.35, "cm"))
  ) +
  coord_sf(ylim = c(-60, 85), expand = FALSE) +
  labs(fill = "Recorded persons") + theme_void() +
  theme(legend.position = "bottom")
```

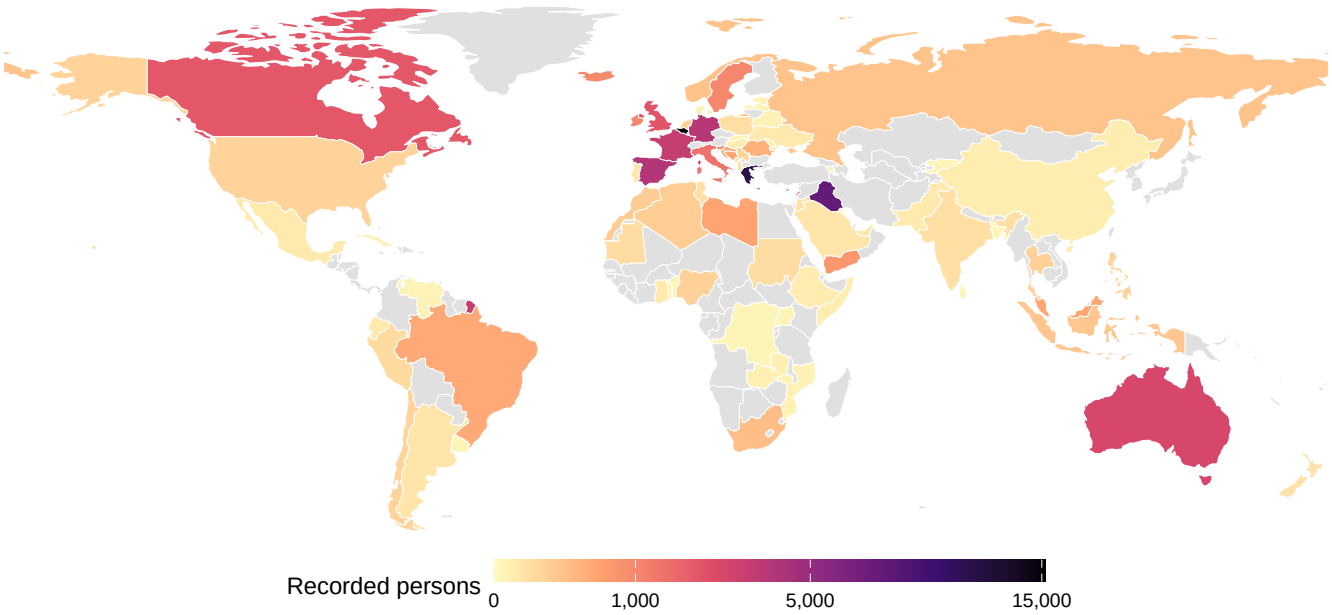


Figure 4: Palestinian-origin recorded populations in 2025. Grey indicates no matched record. Natural Earth boundaries are illustrative.

```
# Figure 5: adapted from the top-ten chart in Part 1.
top10 <- pal25 %>%
  slice_max(total, n = 10) %>%
  mutate(country = if_else(iso3 == "GBR", "United Kingdom", country))
ggplot(top10, aes(total, reorder(country, total))) +
  geom_col(fill = "#287F8A") +
  geom_text(aes(label = scales::comma(total)),
            hjust = -.12, size = 3) +
  scale_x_continuous(labels = scales::comma,
                     expand = expansion(mult = c(0, .16))) +
  labs(x = "Recorded persons", y = NULL)
```

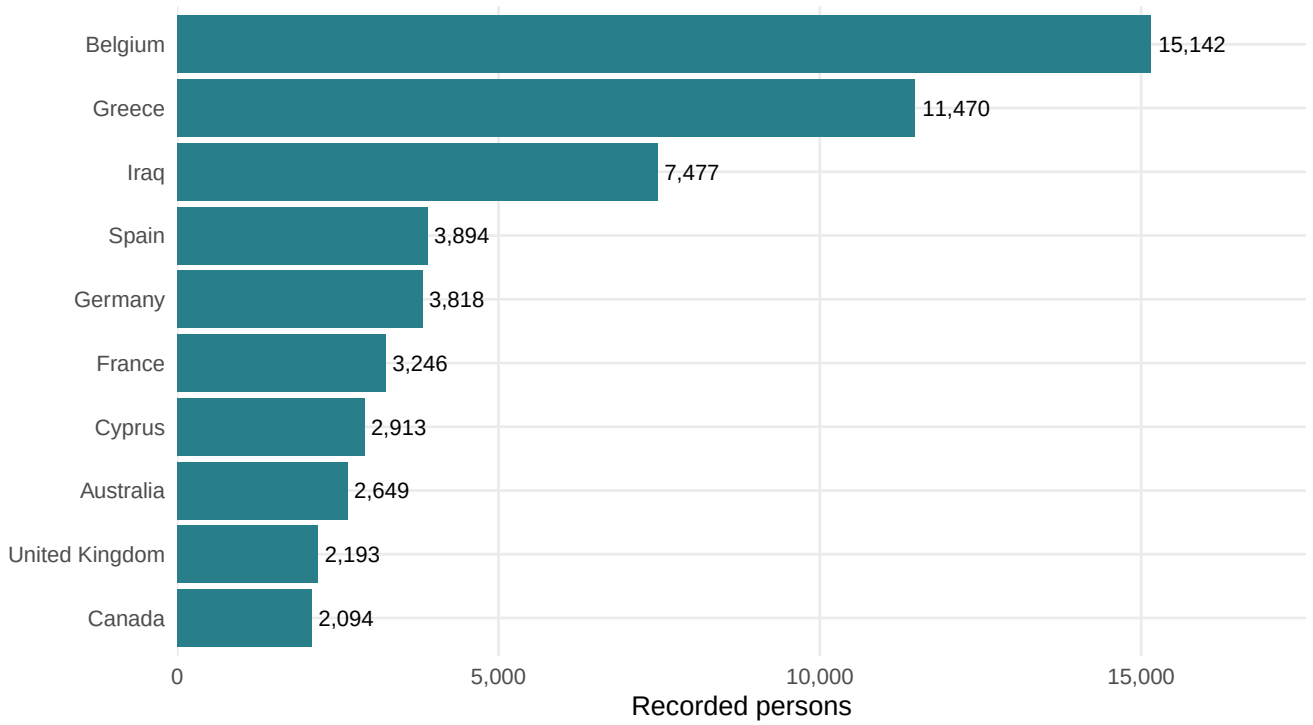


Figure 5: Ten largest Palestinian-origin populations recorded in receiving countries in 2025.

5. Models

a. Motivation behind each model

Model 1 estimates yearly differences using negative-binomial mixed regression: Total depends on categorical Year and a country random intercept. Overdispersed counts motivate NB2; repeated country observations motivate the random intercept. This avoids imposing constant annual growth. Model 2 regresses log(1 + Total) on log GDP per capita and log host population in 2025. It examines economic and population associations while reducing skewness and controlling each predictor for the other.

b. Can the models be compared?

They provide complementary evidence but cannot be directly ranked by fit. Model 1 uses 469 country-year counts; Model 2 uses 77 countries and a transformed response. Their samples, likelihoods and estimands differ.

c. Which model is better, and why?

Model 1 is better suited to longitudinal count variation; Model 2 to cross-sectional associations. Their different outcomes and samples prevent a single fit ranking. A direct comparison would require the same observations, response definition and prediction target.

d. Interpretation in plain English

```
# Model 1: adapted from Part 1 (negative-binomial GLMM).
model_data <- pal %>%
  mutate(Year = factor(Year, levels = 2020:2025),
         Country = factor(`Country of Asylum ISO`))
model1 <- glmmTMB::glmmTMB(
  Total ~ Year + (1 | Country),
  family = glmmTMB::nbinom2(link = "log"),
  data = model_data
)

# Coefficients and exponentiated 95% Wald intervals
c1 <- as.data.frame(summary(model1)$coefficients$cond)
c1$Term <- rownames(c1)
c1$Ratio <- exp(c1$Estimate)
c1$Lower <- exp(c1$Estimate - 1.96 * c1$`Std. Error`)
c1$Upper <- exp(c1$Estimate + 1.96 * c1$`Std. Error`)
c1$`Pr(>|z|)` <- format.pval(c1$`Pr(>|z|)`, digits = 3, eps = .001)
tab(c1 %>% select(Term, Estimate, `Std. Error`, `Pr(>|z|)`,
                  Ratio, Lower, Upper),
    "Model 1 coefficients and exponentiated 95% Wald intervals",
    digits = 4)
```

Table 8: Model 1 coefficients and exponentiated 95% Wald intervals

Term	Estimate	Std. Error	Pr(>|z|)	Ratio	Lower	Upper
(Intercept)	4.0051	0.2395	<0.001	54.8754	34.3202	87.7417
Year2021	0.0552	0.0908	0.5434	1.0567	0.8845	1.2625
Year2022	-0.0212	0.0928	0.8196	0.9791	0.8162	1.1744
Year2023	0.0046	0.0936	0.9607	1.0046	0.8363	1.2069
Year2024	0.1911	0.0940	0.0421	1.2106	1.0068	1.4556
Year2025	0.5876	0.0953	<0.001	1.7998	1.4932	2.1692

Relative to 2020, Model 1 estimates 21.1% higher conditional counts in 2024 ($p = .042$) and 80.0% higher counts in 2025 ($p < .001$). Differences in 2021–2023 are not statistically significant. The estimates account for country heterogeneity within the observed panel.

```r
# Model 2: adapted from Part 4 (country of asylum).
# analysis25 contains the 77 complete country matches shown above.
model2 <- lm(
  log1p(total) ~ log(gdp_pc) + log(host_population),
  data = analysis25
)
c2 <- as.data.frame(summary(model2)$coefficients)
c2$Term <- rownames(c2)
c2$`Pr(>|t|)` <- format.pval(c2$`Pr(>|t|)`, digits = 3, eps = .001)
tab(c2 %>% select(Term, everything()),
    "Model 2 coefficients", digits = 4)
```

Table 9: Model 2 coefficients

Term	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-9.0182	3.1054	-2.9040	0.00485
log(gdp_pc)	0.8402	0.1598	5.2570	< 0.001
log(host_population)	0.3375	0.1315	2.5669	0.01228

Model 2's GDP coefficient is 0.840 ($p < .001$); population's is 0.338 ($p = .012$). Holding the other predictor constant, 1% higher GDP per capita corresponds approximately to 0.84% higher Total + 1; population's equivalent association is 0.34%. Adjusted R^2 is .255, with $F(2,74) = 14.03$. The pattern links higher income and larger populations with higher recorded counts.

Table 10: Separate model summaries

Model	N	Measure	Value
Model 1	469	Random-intercept SD	2.213
Model 2	77	Adjusted R squared	0.255

6. Discussion and conclusion

The increase outside Egypt and Model 1's higher 2024–2025 estimates raise questions about sustained reception needs. In Gaza, OCHA reported approximately 658,000 students without formal schooling and 14,000 patients requiring medical evacuation in December 2024 (OCHA, 2024). These reports connect displacement to disrupted life chances. We interpret continued population growth across receiving countries as a reason to plan education, healthcare and family support over several years, alongside emergency assistance.

Europe's prominence and the positive GDP association suggest that reception capacity deserves attention. Globally, low- and middle-income countries hosted 75% of refugees and other people needing international protection in 2023 (UNHCR, 2024). Our 2025 ranking captures a different segment of this protection geography, shaped partly by UNHCR and UNRWA coverage. International responsibility sharing should therefore support both regional providers and countries receiving onward movement. National wealth becomes relevant through the resources available for protection and settlement.

Model 2's adjusted R^2 of .255 leaves substantial cross-country variation unexplained. Interview research highlights how family reunification, reliable information and social networks shape destination preferences

alongside economic circumstances (Crawley & Hagen-Zanker, 2019). This suggests an interpretation in which resources and social connections work together to make settlement feasible. The male-heavy adult profile makes access to family reunion a particularly relevant question for future research. Belgium's lead over Germany also invites closer examination of recognition practices and established communities.

Research on Palestinian secondary displacement identifies legal protection gaps that become acute when conflict forces renewed flight (Erakat, 2014). This connects our geographical dispersion to a longer history of interrupted security. We conclude that concentrated and growing recorded populations call for durable legal protection, accessible family reunion and sustained public services. Future comparisons should examine how these arrangements translate host-country resources into everyday security, using stable country panels and evidence on refugees' experiences.

Word count: 1,092. Including narrative and section headings; excluding title, in-text citations, references, code and output.

References

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World Bank. (n.d.-b). *Population, total* [Data set]. World Development Indicators. Retrieved November 12, 2026, from https://data.worldbank.org/indicator/SP.POP.TOTL

GitHub Repository

The four R Markdown files containing group members' visualisation code, the consolidated R Markdown report, and the PDF have been uploaded to GitHub:

https://github.com/Costawang/NMC5344-Group-Project-Group7